

Supporting Information

Coverage versus Selectivity in Pre-Export Row Scaling for Generated Mixed-Integer Programs

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This document contains (i) the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text; Figures S1–S3), which mirror the corresponding 1%-MIPGap figures of the main text; (ii) the 0.1%-MIPGap CPLEX distributional and profile views (Scenario 4; Figure S4); (iii) the hyperparameter sensitivity of the structure-aware rule (Table S1); (iv) the per-variant original-scale reliability breakdown with PAR10 under the three readings of the main text (Table S2); (v) the out-of-sample sensitivity of the collective coupling reliability floor (Table S3); (vi) the evaluation of the optional local reliability floor on the failing Simple subset (Table S4); (vii) the common-pool exclusions and full-pool sensitivity (Table S5); (viii) paired-test attrition, the corrected Wilcoxon procedure, and clustered-bootstrap accounting (Table S6); (ix) the available solver-internal-scaling sweep and the verified CPLEX `Read::Scale= -1` audit (Table S7); and (x) the archived runtime comparison used for the coupling-stage activation audit (Table S8); (xi) the direct SA–Flat strict/mixed reliability audit (Table S9); and (xii) the multiplicity, operational-family, bootstrap, preprocessing-plus-solver, deterministic-effort, and own-incumbent sensitivities for that direct comparison, together with its per-contrast attrition and preprocessing-time distributions (Table S10).

S1. Proofs of the spectral propositions

These are the full proofs of the two propositions stated in Section 3 of the main text, moved here to keep the motivation and the method adjacent.

Proof of prop:ideal-block-scaling.

Proof. For fixed positive scalars α_i , the scaled matrix is block diagonal:

$$(\alpha_1 I, \dots, \alpha_B I)D = (\alpha_1 A_1, \dots, \alpha_B A_B).$$

The singular values of a block-diagonal matrix are the union of the singular values of its blocks. Therefore, for every i ,

$$\kappa_2((\alpha_1 A_1, \dots, \alpha_B A_B)) \geq \kappa_2(\alpha_i A_i) = \kappa_2(A_i).$$

Taking the maximum over i gives the lower bound

$$\kappa_2((\alpha_1 A_1, \dots, \alpha_B A_B)) \geq \max_i \kappa_2(A_i)$$

for every admissible block scaling.

Now choose $\alpha_i = (M_i m_i)^{-1/2}$. Then the largest and smallest singular values of $\alpha_i A_i$ are

$$\begin{aligned}\sigma_{\max}(\alpha_i A_i) &= \sqrt{\frac{M_i}{m_i}} = \sqrt{\kappa_2(A_i)}, \\ \sigma_{\min}(\alpha_i A_i) &= \sqrt{\frac{m_i}{M_i}} = \frac{1}{\sqrt{\kappa_2(A_i)}}.\end{aligned}$$

Because the singular values of the complete block-diagonal matrix are obtained by pooling the singular values of its blocks,

$$\sigma_{\max}((\alpha_1 A_1, \dots, \alpha_B A_B)) = \sqrt{\max_i \kappa_2(A_i)},$$

and

$$\sigma_{\min}((\alpha_1 A_1, \dots, \alpha_B A_B)) = \frac{1}{\sqrt{\max_i \kappa_2(A_i)}}.$$

The resulting condition number is $\max_i \kappa_2(A_i)$, matching the lower bound. $\square$

Proof of prop:triangular-coupling.

Proof. Let $R = U\Sigma V^\top$ be a singular value decomposition (SVD) of R , with U and V orthogonal. Orthogonal multiplication does not change singular values, and hence T_R has the same singular values as

$$\begin{bmatrix} V^\top & 0 \\ 0 & U^\top \end{bmatrix} \begin{bmatrix} I_n & 0 \\ R & I_m \end{bmatrix} \begin{bmatrix} V & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ \Sigma & I_m \end{bmatrix}.$$

For each singular value σ of R , the nontrivial part reduces to the two-dimensional block

$$B_\sigma = \begin{bmatrix} 1 & 0 \\ \sigma & 1 \end{bmatrix}.$$

The squared singular values of B_σ are the eigenvalues of

$$B_\sigma^\top B_\sigma = \begin{bmatrix} 1 + \sigma^2 & \sigma \\ \sigma & 1 \end{bmatrix},$$

namely

$$\lambda_\pm(\sigma) = \frac{\sigma^2 + 2 \pm \sigma\sqrt{\sigma^2 + 4}}{2}.$$

Their product is one, and $\lambda_+(\sigma)$ is increasing for $\sigma \geq 0$. Additional directions associated with zero singular values of R contribute singular values equal to one. Therefore the largest squared singular value of T_R is $\lambda_+(\rho)$ and the smallest squared singular value is $1/\lambda_+(\rho)$. Consequently,

$$\begin{aligned} \kappa_2(T_R) &= \lambda_+(\rho) \\ &= \frac{\rho^2 + 2 + \rho\sqrt{\rho^2 + 4}}{2} \\ &= \left(\frac{\sqrt{\rho^2 + 4} + \rho}{2} \right)^2. \end{aligned}$$

$\square$

Here, PAR10 (including its ITT/strict/mixed superscripts) denotes the common-pool metric defined in the main text. The distinct symbol $\text{PAR10}_{\text{avail}}$ is reserved for archived descriptive revision tables: its denominator is the variant's available solver-output records, a timeout receives 36,000 s, and a pure pipeline error with no solver output is excluded. It must not be compared numerically with the fixed-pool metric.

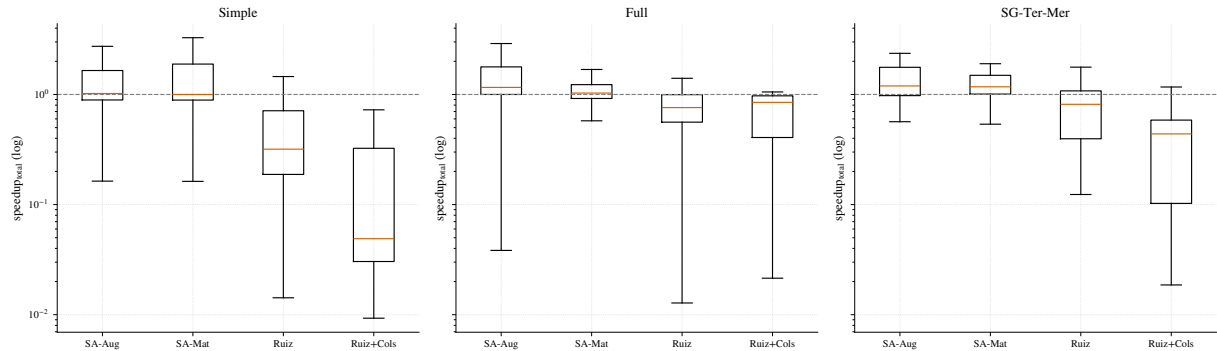
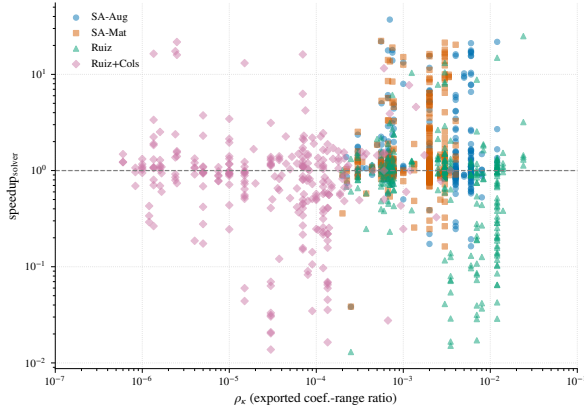
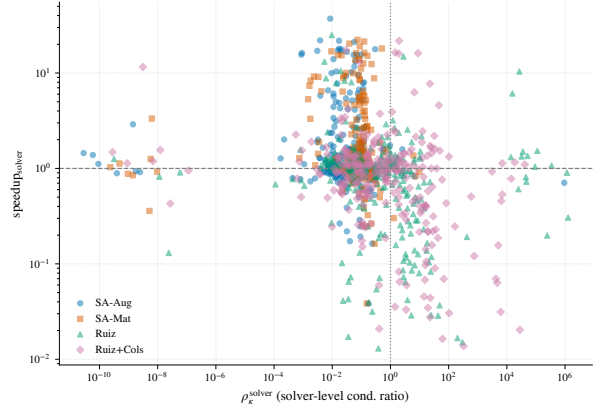


Figure S1: At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.



(a) Static coefficient-range ratio.



(b) Solver-level conditioning ratio.

Figure S2: At 0.1% MIPGap, the pooled scatter shows a stronger association with speedup for the own-incumbent path-specific fixed-LP diagnostic than for the exported-range proxy. This describes solver paths; it does not make the former a purer measure of representation or an instance-level predictor for the structure-aware variants.

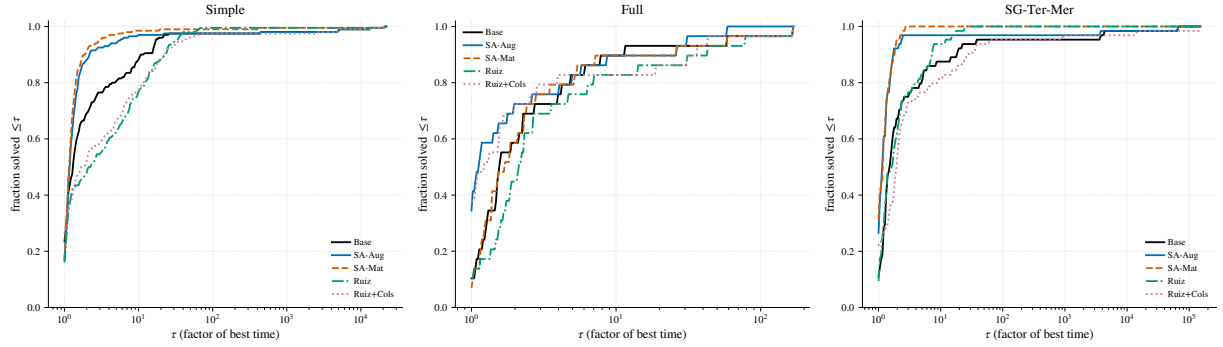
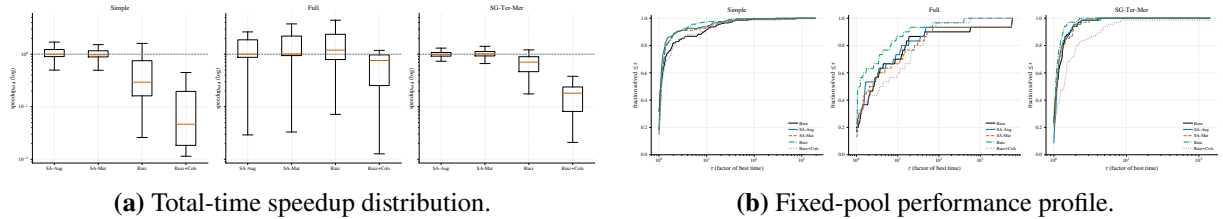


Figure S3: At 0.1% MIPGap, fixed-pool Dolan–Moré profiles use the common denominators and individual 36,000-s failure cost defined in the main text.



(a) Total-time speedup distribution.

(b) Fixed-pool performance profile.

Figure S4: At 0.1% MIPGap, the CPLEX visual evidence remains mixed under the stricter tolerance. The profile uses a common pool and assigns 36,000 s to an individual timeout, abort, missing run, or invalid time, as defined in the main text.

Table S1: Hyperparameter sensitivity of SA-Aug (Simple class, Gurobi, 1% MIPGap). Solver-time sgm ($s = 1$ s) and descriptive $\text{PAR10}_{\text{avail}}$, in seconds; default settings in bold.

Parameter	Value	sgm (s)	$\text{PAR10}_{\text{avail}}$ (s)
Band u	1.5	2.09	2.46
	2	2.07	2.45
	4	2.12	2.49
Window $[\varepsilon_{\min}, \varepsilon_{\max}]$	$[10^{-6}, 10^6]$	2.04	2.40
	$[10^{-9}, 10^9]$	2.02	2.39
	$[10^{-12}, 10^{12}]$	2.01	2.36

Base sgm is 3.73 s for reference. Each row is a separate matched run; the spread across all settings ($\text{sgm} \in [2.01, 2.12]$) is far below the base-to-scaled gap, and the median paired speedup is ≈ 1.0 throughout. The number of Ruiz iterations K is a parameter of the Ruiz baselines, not of the structure-aware rule, so it is fixed at $K = 4$; varying it leaves SA-Aug unchanged, as expected.

Table S2: Per-variant audit and PAR10 under the three operational readings. All campaigns use the manifest-declared $T_{\max} = 3600$ s. A timeout or pipeline abort receives $10T_{\max} = 36000$ s in every reading; a returned output that fails the relevant verifier receives that penalty only in the strict or mixed reading. The count columns use the raw manifest pool: n_{att} is every attempted instance and n_{out} reproduces the pooled diagnostic audit’s status-OK denominator, including time-limited runs and three records that lack verification fields. TO is a subset of n_{out} , and Abort is $n_{\text{att}} - n_{\text{out}}$. Pass counts are among those status-OK records; missing verification is non-passing. Strict pass means $v_{\text{row}} \leq 10^{-4}$ and $v_{\text{int}} \leq 10^{-5}$; mixed pass rescues only row failures with $v_r/(1 + |b_r|) \leq 10^{-6}$. In contrast, the four PAR10 columns use the reproducible scenario-specific common pool after the uniformly-unavailable exclusion in Table S5: Base ITT and, for the listed variant, ITT/strict/mixed. Values are seconds, rounded to the nearest second.

Solver	Gap	Class	Variant	n_{att}	n_{out}	TO	Abort	$n_{\text{pass}}^{\text{str}}$	$n_{\text{pass}}^{\text{mix}}$	Base ITT	ITT	Strict	Mixed
Gurobi	1%	Simple	SA-Aug	204	197	0	7	185	185	373	364	2535	2535
Gurobi	1%	Simple	SA-Mat	204	198	0	6	179	180	373	183	3620	3439
Gurobi	1%	Simple	Ruiz	204	198	0	6	143	146	373	207	10153	9611
Gurobi	1%	Simple	Ruiz+Cols	204	199	0	5	180	189	373	28	3462	1835
Gurobi	1%	SG-Ter-Mer	SA-Aug	68	67	1	1	66	67	1094	1083	1613	1083
Gurobi	1%	SG-Ter-Mer	SA-Mat	68	67	0	1	65	66	1094	1079	1608	1079
Gurobi	1%	SG-Ter-Mer	Ruiz	68	68	1	0	67	68	1094	567	1095	567
Gurobi	1%	SG-Ter-Mer	Ruiz+Cols	68	68	1	0	67	68	1094	585	1113	585
Gurobi	1%	Full	SA-Aug	34	34	3	0	27	27	3385	3308	9640	9640
Gurobi	1%	Full	SA-Mat	34	34	2	0	31	31	3385	3473	5586	5586
Gurobi	1%	Full	Ruiz	34	33	4	1	31	31	3385	5491	7577	7577
Gurobi	1%	Full	Ruiz+Cols	34	33	4	1	31	32	3385	5617	6666	5617
Gurobi	0.1%	Simple	SA-Aug	204	195	0	9	174	178	928	928	4705	3987
Gurobi	0.1%	Simple	SA-Mat	204	198	0	6	168	172	928	381	5776	5057
Gurobi	0.1%	Simple	Ruiz	204	199	0	5	155	159	928	221	8138	7419
Gurobi	0.1%	Simple	Ruiz+Cols	204	195	0	9	175	185	928	948	4544	2746
Gurobi	0.1%	SG-Ter-Mer	SA-Aug	68	63	1	5	62	63	2246	1683	2237	1683
Gurobi	0.1%	SG-Ter-Mer	SA-Mat	68	65	0	3	63	64	2246	576	1130	576
Gurobi	0.1%	SG-Ter-Mer	Ruiz	68	65	1	3	65	65	2246	587	587	587
Gurobi	0.1%	SG-Ter-Mer	Ruiz+Cols	68	62	1	6	61	62	2246	2262	2814	2262
Gurobi	0.1%	Full	SA-Aug	34	30	3	4	27	27	5992	5897	8140	8140
Gurobi	0.1%	Full	SA-Mat	34	31	4	3	28	30	5992	6026	9293	7148
Gurobi	0.1%	Full	Ruiz	34	30	4	4	27	27	5992	7066	9277	9277
Gurobi	0.1%	Full	Ruiz+Cols	34	31	5	3	28	29	5992	7073	9312	8191
CPLEX	1%	Simple	SA-Aug	204	204	0	0	37	37	6	3	29471	29471
CPLEX	1%	Simple	SA-Mat	204	204	0	0	43	43	6	3	28412	28412
CPLEX	1%	Simple	Ruiz	204	204	0	0	44	44	6	4	28237	28237
CPLEX	1%	Simple	Ruiz+Cols	204	204	0	0	137	138	6	3	11825	11649
CPLEX	1%	SG-Ter-Mer	SA-Aug	68	68	0	0	68	68	3	10	10	10
CPLEX	1%	SG-Ter-Mer	SA-Mat	68	68	0	0	68	68	3	6	6	6
CPLEX	1%	SG-Ter-Mer	Ruiz	68	68	0	0	68	68	3	10	10	10
CPLEX	1%	SG-Ter-Mer	Ruiz+Cols	68	68	0	0	68	68	3	15	15	15

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Table S2 (continued)

Solver	Gap	Class	Variant	n_{att}	n_{out}	TO	Abort	$n_{\text{pass}}^{\text{str}}$	$n_{\text{pass}}^{\text{mix}}$	Base ITT	ITT	Strict	Mixed
CPLEX	1%	Full	SA-Aug	34	34	5	0	25	25	6515	5546	15014	15014
CPLEX	1%	Full	SA-Mat	34	34	6	0	25	26	6515	6498	14932	13875
CPLEX	1%	Full	Ruiz	34	34	3	0	28	28	6515	3506	8744	8744
CPLEX	1%	Full	Ruiz+Cols	34	34	4	0	32	33	6515	4681	6793	5737
CPLEX	0.1%	Simple	SA-Aug	204	204	1	0	40	40	214	188	29119	29119
CPLEX	0.1%	Simple	SA-Mat	204	204	1	0	38	38	214	203	29311	29311
CPLEX	0.1%	Simple	Ruiz	204	204	1	0	44	44	214	192	28413	28413
CPLEX	0.1%	Simple	Ruiz+Cols	204	204	3	0	144	146	214	547	10955	10779
CPLEX	0.1%	SG-Ter-Mer	SA-Aug	68	68	0	0	68	68	19	21	21	21
CPLEX	0.1%	SG-Ter-Mer	SA-Mat	68	68	0	0	68	68	19	16	16	16
CPLEX	0.1%	SG-Ter-Mer	Ruiz	68	68	0	0	68	68	19	8	8	8
CPLEX	0.1%	SG-Ter-Mer	Ruiz+Cols	68	68	1	0	68	68	19	539	539	539
CPLEX	0.1%	Full	SA-Aug	34	34	8	0	24	24	9951	8930	18283	18283
CPLEX	0.1%	Full	SA-Mat	34	33	10	1	25	26	9951	11916	20280	19226
CPLEX	0.1%	Full	Ruiz	34	34	5	0	32	32	9951	5857	7917	7917
CPLEX	0.1%	Full	Ruiz+Cols	34	33	10	1	31	32	9951	11950	14063	13007

Objective-difference audit. MIPGap bounds each incumbent against its own dual bound and is not a cross-run agreement test. Of 4,806 variant runs with returned solver output, 72 differ from the corresponding Base incumbent by more than twice the nominal gap; these cases are concentrated in a small set of instance identifiers. The largest pooled relative deviations occur in Full/Ruiz cells ($|\Delta z_{\text{rel}}| \approx 1.8 \times 10^2$ under Gurobi and 5.3×10^1 under CPLEX). Base is not presumed correct: one Simple Base incumbent is 73% worse than the best verified incumbent, and one CPLEX Full Base incumbent is roughly six times worse. Accordingly, correctness is assessed against the best original-scale-verified incumbent on each instance, with Base subjected to the same feasibility and integrality checks as every scaled variant.

Table S3: Out-of-sample sensitivity of the reliability floor for the collective coupling factor (SA-Aug), on twenty seeds (21–40) never used while designing the floor. Cells: coupling, coupling-uniform, and mixed patterns at severity $S = 6$ (60 instances, Gurobi, MIPGap 10^{-3} , 600 s limit). Median post-scaling global coefficient-and-right-hand-side magnitude-range proxy per pattern; number of runs failing original-scale verification; worst original-scale row violation; instances whose objective disagrees with the base beyond the MIPGap. The failure mode replicates out-of-sample with the floor disabled, and every floor value across three orders of magnitude eliminates it.

Floor	coupling	coupling-unif.	mixed	n_{fail}	v_{max}	over gap
(pre-scaling)	4.4×10^8	3.3×10^6	1.5×10^{14}			
disabled	3.8×10^7	5.4×10^3	2.9×10^7	4	4.2×10^{-1}	4
10^{-2}	9.1×10^5	5.4×10^3	2.9×10^6	0	1.2×10^{-7}	0
10^{-3} (default)	3.5×10^5	5.4×10^3	2.3×10^6	0	1.2×10^{-7}	0
10^{-4}	1.0×10^6	5.4×10^3	2.3×10^6	0	1.2×10^{-7}	0
10^{-5}	2.1×10^6	5.4×10^3	3.0×10^6	0	1.2×10^{-7}	0

Coupling-trigger sensitivity. On the original synthetic grid, sweeping $\hat{\rho}_{\text{trigger}} \in \{2, 5, 10, 20, 50\}$ leaves every displayed coupling and coupling-uniform cell median unchanged. Approximately one instance in twenty lies in the $[2, 50]$ band, so some individual activation decisions change, but the proxy is effectively bimodal (above 10^2 on oversized instances and below one on undersized instances). The result therefore supports the stated conclusion about the one-sided design, not a privileged choice of the cutoff 10.

Table S4: Optional local reliability floor (SA-Aug-LF), evaluated on the twelve Simple instances (Gurobi, 1% MIPGap) whose SA-Aug runs fail original-scale verification. The floor applies the local factor only if $\alpha_r^{A,b} \geq \varepsilon_{\text{floor}}^{\text{loc}}$, so $d_r \geq \varepsilon_{\text{floor}}^{\text{loc}}$ bounds the conditional original-scale residual by $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$; rows below the floor stay at base scale. n_{fail} is the number of the twelve runs failing the strict absolute criterion; σ_{gm} is the shifted geometric mean solver time ($s = 1$ s); $\hat{\kappa}_{\text{range}}^{\text{post}}$ is the median exported coefficient-and-right-hand-side magnitude-range proxy; v_{max} is the worst original-scale row violation. SA-Mat is shown for reference.

Variant	n_{fail}	σ_{gm} (s)	$\hat{\kappa}_{\text{range}}^{\text{post}}$	v_{max}
Base	0	4.90	—	0
SA-Aug (no floor, default)	12	2.47	6.6×10^9	3.7×10^{-3}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-1}$	0	2.64	5.5×10^{11}	5.3×10^{-5}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-2}$	0	2.89	5.5×10^{11}	7.4×10^{-5}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-3}$	6	5.52	2.0×10^{14}	4.0×10^{-2}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-4}$	6	2.19	2.0×10^{14}	5.3×10^{-4}
SA-Mat	5	2.57	4.5×10^{12}	1.9×10^{-3}

Notes. The floor is monotone: n_{fail} falls from 12 (no floor) to 6 (shallow floors $10^{-3}/10^{-4}$, which protect only the most extreme rows) to 0 ($10^{-2}/10^{-1}$), consistent with the residual bound $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$. At the effective floor 10^{-2} the variant eliminates every failure while retaining a solver-time advantage over the base (σ_{gm} 2.89 vs 4.90 s), even though it *raises* the exported magnitude-range proxy (the protected rows are exactly the over-compressed large-right-hand-side rows)—a further illustration that this static proxy is a weak predictor of runtime. The threshold is imposed on the representation-dependent factor $\alpha_r^{A,b}$ and is therefore not invariant to a positive re-emission of the same unscaled row. Here it is defined relative to the generator’s canonical export in physical units. A representation-invariant alternative could threshold a norm of the resulting scaled row, but it would not inherit the same $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$ residual bound. The σ_{gm} values on twelve instances are noisy and are not a full-class estimate. This subset was selected by a post-treatment outcome (the SA-Aug failures), so the table is a demonstration on the failing set; the full-class reliability and runtime effect, and the behavior on Full, remain untested.

Table S5: Uniformly unavailable operational records and sensitivity to retaining them. Panel A applies the documented reproducible data rule: exclude an instance only when all five attempted variants have nonzero pipeline return codes. Panel B restores those cells to the manifest-declared full pool and assigns each a 36000-s penalty. Triplets are ITT/strict/mixed PAR10 in seconds; only scenarios with exclusions are shown because common- and full-pool results otherwise coincide. The within-reading ordering of Base, SA-Aug, and SA-Mat is unchanged (the rounded common-pool tie in Gurobi Simple at 0.1% is broken by less than one second before rounding in the full pool).

Panel A: identifiers				
Solver	Gap	Class	N/n	Uniformly unavailable identifiers
Gurobi	1%	Simple	204/199	caso46f, caso47a–caso47d
Gurobi	0.1%	Simple	204/200	caso02c–caso02e, caso22a
Gurobi	0.1%	SG-Ter-Mer	68/65	instance020–instance022
Gurobi	0.1%	Full	34/32	instance046, instance047

Panel B: common pool versus penalized full pool

Gap/class	Variant	Pool	ITT	Strict	Mixed
1%, Simple ($n = 199$, $N = 204$)	Base	common	373	554	554
		full	1246	1422	1422
	SA-Aug	common	364	2535	2535
		full	1238	3355	3355
	SA-Mat	common	183	3620	3439
		full	1061	4414	4237
	Ruiz	common	207	10153	9611
		full	1085	10786	10258
	Ruiz+Cols	common	28	3462	1835
		full	909	4260	2673
0.1%, Simple ($n = 200$, $N = 204$)	Base	common	928	1108	1108
		full	1615	1792	1792
	SA-Aug	common	928	4705	3987
		full	1616	5318	4615
	SA-Mat	common	381	5776	5057
		full	1079	6369	5664
	Ruiz	common	221	8138	7419
		full	923	8684	7980
	Ruiz+Cols	common	948	4544	2746
		full	1635	5161	3398
0.1%, SG-Ter-Mer ($n = 65$, $N = 68$)	Base	common	2246	2246	2246
		full	3735	3735	3735
	SA-Aug	common	1683	2237	1683
		full	3197	3727	3197
	SA-Mat	common	576	1130	576
		full	2139	2668	2139
	Ruiz	common	587	587	587
		full	2149	2149	2149
	Ruiz+Cols	common	2262	2814	2262
		full	3750	4279	3750
0.1%, Full ($n = 32$, $N = 34$)	Base	common	5992	5992	5992
		full	7757	7757	7757
	SA-Aug	common	5897	8140	8140
		full	7668	9779	9779
	SA-Mat	common	6026	9293	7148
		full	7789	10864	8845
	Ruiz	common	7066	9277	9277
		full	8768	10849	10849
	Ruiz+Cols	common	7073	9312	8191
		full	8775	10882	9827

Note. N is the manifest pool and n the common pool. In the archived logs for every excluded identifier, Base, SA-Aug, Ruiz, and Ruiz+Cols stop at a Gurobi WLS `Overage for too long` licensing error and report no configured solver. The SA-Mat logs instead record optimal completions written to a separate `*-matricial` result tree, while their aggregate campaign records retain nonzero return codes. Thus the evidence supports a combination of license-capacity failure and incomplete artifact consolidation; the exclusion itself is defined only by the observable all-variant return-code rule and does not assert that every underlying solve failed.

Table S6: Pair attrition and corrected Wilcoxon audit. n is the effective matched-pair count. UM denotes the all-variant uniformly-unavailable exclusion; D_{TO} , D_A , and D_{mix} are exclusions because both Base and the listed variant are, respectively, timeouts, pipeline aborts, or different kinds of noncompletion. C_1 is a one-sided noncompletion retained at exactly $T_{max} = 3600$ s. There are no missing-time or exact-zero pairs in any row. A one-sided timeout is right-censored at T_{max} ; assigning the same cost to a one-sided algorithmic/pipeline abort is an operational-failure convention rather than a censoring claim. The test is two-sided Wilcoxon signed-rank on $t_{variant} - t_{Base}$ (SciPy 1.15.3), with exact zeros discarded, tied absolute differences assigned midranks, and no continuity correction. The exact null distribution is used for at most 50 nonzero untied differences; otherwise the tie-corrected normal approximation is used. The displayed structure-aware Simple/SG rows use the approximation and Full rows use the exact distribution; no zero or tied-absolute-difference pair occurs. $r_{rb} < 0$ favors the listed variant. q is Benjamini–Hochberg adjusted over the twelve tests in each solver-by-gap family (three classes by four variants).

Solver	Gap	Class	Variant	n	UM	D_{TO}	D_A	D_{mix}	C_1	W	p	q	r_{rb}
Gurobi	1%	Simple	SA-Aug	197	5	0	2	0	0	7535	5.67×10^{-3}	9.72×10^{-3}	−0.227
Gurobi	1%	Simple	SA-Mat	198	5	0	1	0	1	3011	2.42×10^{-17}	2.90×10^{-16}	−0.694
Gurobi	1%	Simple	Ruiz	198	5	0	1	0	1	7248	1.27×10^{-3}	3.04×10^{-3}	+0.264
Gurobi	1%	Simple	Ruiz+Cols	199	5	0	0	0	2	7414	1.82×10^{-3}	3.65×10^{-3}	+0.255
Gurobi	1%	SG-Ter-Mer	SA-Aug	66	0	1	1	0	0	501	1.13×10^{-4}	4.51×10^{-4}	−0.547
Gurobi	1%	SG-Ter-Mer	SA-Mat	66	0	0	1	1	0	115	2.49×10^{-10}	1.50×10^{-9}	−0.896
Gurobi	1%	SG-Ter-Mer	Ruiz	67	0	1	0	0	1	1129	0.950	0.950	−0.009
Gurobi	1%	SG-Ter-Mer	Ruiz+Cols	67	0	1	0	0	1	598	7.26×10^{-4}	2.18×10^{-3}	+0.475
Gurobi	1%	Full	SA-Aug	32	0	2	0	0	2	175	0.0984	0.148	−0.337
Gurobi	1%	Full	SA-Mat	32	0	2	0	0	2	204	0.270	0.360	−0.227
Gurobi	1%	Full	Ruiz	31	0	2	0	1	2	197	0.327	0.393	+0.206
Gurobi	1%	Full	Ruiz+Cols	32	0	2	0	0	4	243	0.705	0.769	+0.080
Gurobi	0.1%	Simple	SA-Aug	195	4	0	5	0	0	6963	1.02×10^{-3}	3.06×10^{-3}	−0.271
Gurobi	0.1%	Simple	SA-Mat	198	4	0	2	0	3	7069	5.70×10^{-4}	2.28×10^{-3}	−0.282
Gurobi	0.1%	Simple	Ruiz	199	4	0	1	0	4	7429	1.94×10^{-3}	4.66×10^{-3}	+0.253
Gurobi	0.1%	Simple	Ruiz+Cols	199	4	0	1	0	8	7517	2.78×10^{-3}	5.46×10^{-3}	+0.245
Gurobi	0.1%	SG-Ter-Mer	SA-Aug	62	3	1	2	0	1	456	2.63×10^{-4}	1.58×10^{-3}	−0.533
Gurobi	0.1%	SG-Ter-Mer	SA-Mat	64	3	0	0	1	3	341	2.95×10^{-6}	3.53×10^{-5}	−0.672
Gurobi	0.1%	SG-Ter-Mer	Ruiz	64	3	1	0	0	3	889	0.313	0.341	−0.145
Gurobi	0.1%	SG-Ter-Mer	Ruiz+Cols	64	3	1	0	0	6	599	3.19×10^{-3}	5.46×10^{-3}	+0.424
Gurobi	0.1%	Full	SA-Aug	28	2	3	1	0	2	124	0.0735	0.110	−0.389
Gurobi	0.1%	Full	SA-Mat	27	2	4	1	0	0	173	0.714	0.714	−0.085
Gurobi	0.1%	Full	Ruiz	28	2	3	0	1	3	144	0.186	0.248	+0.291
Gurobi	0.1%	Full	Ruiz+Cols	28	2	4	0	0	3	149	0.227	0.272	−0.266
CPLEX	1%	Simple	SA-Aug	204	0	0	0	0	0	9377	0.202	0.346	−0.103
CPLEX	1%	Simple	SA-Mat	204	0	0	0	0	0	7460	3.89×10^{-4}	2.33×10^{-3}	+0.286
CPLEX	1%	Simple	Ruiz	204	0	0	0	0	0	9496	0.256	0.384	+0.092
CPLEX	1%	Simple	Ruiz+Cols	204	0	0	0	0	0	8958	0.0762	0.183	+0.143
CPLEX	1%	SG-Ter-Mer	SA-Aug	68	0	0	0	0	0	1068	0.521	0.521	+0.090
CPLEX	1%	SG-Ter-Mer	SA-Mat	68	0	0	0	0	0	740	8.15×10^{-3}	0.0326	+0.369

Continued on next page

Table S6 (continued)

Solver	Gap	Class	Variant	n	UM	D_{TO}	D_A	D_{mix}	C_1	W	p	q	r_{tb}
CPLEX	1%	SG-Ter-Mer	Ruiz	68	0	0	0	0	0	1041	0.420	0.458	+0.113
CPLEX	1%	SG-Ter-Mer	Ruiz+Cols	68	0	0	0	0	0	438	7.09×10^{-6}	8.50×10^{-5}	+0.627
CPLEX	1%	Full	SA-Aug	29	0	4	0	1	1	172	0.336	0.428	+0.209
CPLEX	1%	Full	SA-Mat	28	0	5	0	1	0	131	0.104	0.208	−0.355
CPLEX	1%	Full	Ruiz	31	0	2	0	1	3	200	0.357	0.428	−0.194
CPLEX	1%	Full	Ruiz+Cols	30	0	3	0	1	2	130	0.0345	0.104	+0.441
CPLEX	0.1%	Simple	SA-Aug	204	0	0	0	0	2	8765	0.0453	0.121	−0.162
CPLEX	0.1%	Simple	SA-Mat	203	0	1	0	0	0	10056	0.723	0.723	−0.029
CPLEX	0.1%	Simple	Ruiz	204	0	0	0	0	2	9364	0.196	0.356	−0.104
CPLEX	0.1%	Simple	Ruiz+Cols	203	0	1	0	0	2	8092	6.98×10^{-3}	0.0419	+0.218
CPLEX	0.1%	SG-Ter-Mer	SA-Aug	68	0	0	0	0	0	1085	0.591	0.709	+0.075
CPLEX	0.1%	SG-Ter-Mer	SA-Mat	68	0	0	0	0	0	985	0.251	0.372	−0.160
CPLEX	0.1%	SG-Ter-Mer	Ruiz	68	0	0	0	0	0	853	0.0505	0.121	−0.273
CPLEX	0.1%	SG-Ter-Mer	Ruiz+Cols	68	0	0	0	0	1	473	1.89×10^{-5}	2.27×10^{-4}	+0.597
CPLEX	0.1%	Full	SA-Aug	27	0	6	0	1	3	173	0.714	0.723	−0.085
CPLEX	0.1%	Full	SA-Mat	26	0	6	0	2	4	125	0.208	0.356	−0.288
CPLEX	0.1%	Full	Ruiz	30	0	3	0	1	6	134	0.0427	0.121	−0.424
CPLEX	0.1%	Full	Ruiz+Cols	27	0	6	0	1	6	143	0.279	0.372	+0.243

II

One-sided-abort sensitivity. Replacing the primary $T_{\max}$ convention by $10T_{\max}$ for a one-sided algorithmic/pipeline abort leaves every effect direction and inferential decision unchanged. The largest change among the displayed Gurobi structure-aware rows is Full/SA-Aug at 0.1%: $W = 124 \rightarrow 125$, $p = 0.07354 \rightarrow 0.07742$, $q_{BH} = 0.11031 \rightarrow 0.11612$, and $r_{tb} = -0.389 \rightarrow -0.384$. All other Gurobi structure-aware rows are unchanged; CPLEX structure-aware W , p , and r_{tb} are unchanged, with only minor BH shifts caused by other members of the twelve-test family. The sensitivity is reproduced by `scripts/paired_tests.py --abort-penalty par10`.

Cluster-bootstrap accounting for the conditioning comparison. At 1%, 298 of 306 instances have at least one complete conditioning–speedup pair; eight are lost because Base errored. The whole-instance bootstrap uses 1185 complete variant–instance observations (seven of the possible 298×4 lack the conditioning diagnostic). At 0.1%, 288 instances contribute; eighteen Base errors remove nine Simple, six SG-Ter-Mer, and three Full instances. Of 1152 potential rows, 1135 enter the joint comparison: twelve lack a usable variant speedup and five more lack the solver-conditioning diagnostic. Each of the 2000 resamples keeps all available formulation rows of an instance together. At 1%, the within-class solver-level coefficients are -0.31 (Simple), -0.13 (SG-Ter-Mer), and -0.12 (Full); their varying magnitudes reinforce that the pooled coefficient is descriptive rather than an instance-level predictor.

Exploratory dependence check. If repeated rows and seeds are naively treated as independent observations, the first-order partial rank correlation between runtime improvement and the solver-facing conditioning diagnostic is approximately -0.25 after controlling for loaded nonzeros, and a Williams–Steiger comparison gives $Z \approx 6.1$. Because both calculations ignore the clustered generator design, they are reported only as descriptive diagnostics and support no confirmatory claim.

Table S7: Available solver-internal-scaling sweep over both solvers, both MIPGap settings, and all three classes. Each cell is solver-time $\text{sgm} (s = 1 \text{ s})/\text{PAR10}_{\text{avail}}$ in seconds, using the descriptive revision-campaign convention (timeouts receive 36000 s; pure pipeline errors are excluded). SA-Aug and SA-Mat retain the solver default. For Gurobi, Off/A/B/C mean $\text{ScaleFlag}=0/1/2/3$. For CPLEX they mean $\text{Read}::\text{Scale}=-1/0/1/\text{not applicable}$. The CPLEX Full 0.1% $\text{Scale}=0$ artifact is partial (26 of 34 rows), and $\text{Scale}=1$ is absent because the four-day sweep budget expired; all other configuration folders contain the manifest-declared number of rows. The lower panel audits the apparently fast CPLEX $\text{Scale}=-1$ Full cell directly from the stored original-scale verifier fields.

Panel A: timing sweep ($\text{sgm}/\text{PAR10}_{\text{avail}}$)									
Solver	Gap	Class	Base	SA-Aug	SA-Mat	Off	A	B	C
Gurobi	1%	Simple	3.936/11.26	2.133/2.50	2.166/2.56	41.969/1541.92	5.811/20.39	4.402/12.60	3.791/10.79
Gurobi	1%	SG-Ter-Mer	14.087/564.33	8.282/554.10	8.635/554.64	56.625/740.17	14.664/566.35	20.907/589.78	21.475/580.46
Gurobi	1%	Full	109.375/3448.17	95.239/3349.65	102.025/3458.10	561.945/9716.08	128.449/2444.67	138.112/2497.10	133.828/3468.97
Gurobi	0.1%	Simple	9.656/36.16	6.419/210.83	6.248/202.17	89.358/2031.10	15.299/219.91	10.607/33.11	9.438/35.10
Gurobi	0.1%	SG-Ter-Mer	14.587/566.63	8.811/554.42	9.066/555.19	54.784/727.17	16.972/620.44	11.753/32.85	23.102/585.62
Gurobi	0.1%	Full	437.243/4766.52	388.446/4714.38	456.365/4821.25	396.978/9577.84	337.331/4621.00	347.613/3687.08	353.389/4697.74
CPLEX	1%	Simple	2.621/7.24	1.725/3.65	1.861/4.01	2.500/3.93	2.549/7.59	2.431/7.18	–
CPLEX	1%	SG-Ter-Mer	2.253/3.57	2.400/10.26	2.309/5.94	5.168/12.29	2.229/3.53	2.175/5.66	–
CPLEX	1%	Full	147.821/5609.27	127.360/3616.84	120.881/6509.40	14.381/35.45	159.012/5621.84	189.129/5644.99	–
CPLEX	0.1%	Simple	5.143/215.88	4.266/188.91	4.450/203.68	7.634/209.70	5.203/216.59	5.625/369.68	–
CPLEX	0.1%	SG-Ter-Mer	2.664/22.29	2.678/24.19	2.667/19.50	5.844/13.69	2.725/23.13	2.477/11.78	–
CPLEX	0.1%	Full	425.043/9419.41	426.712/9179.82	387.204/11235.11	13.476/24.97	303.376/6196.93 [†]	–	–

Panel B: CPLEX Full, $\text{Read}::\text{Scale} = -1$, original-scale audit							
Gap	Runs	Strict pass	Mixed pass	$\max v_{\text{row}}$	$\max v_{\text{int}}$	$\max z_{\text{solver}} - z_{\text{recomputed}} $	Strict-fail identifiers
1%	34	26	26	1.3149×10^{-2}	0	3.73×10^{-8}	002,003,004,012,014,022,023,044
0.1%	34	26	26	1.3149×10^{-2}	0	3.73×10^{-8}	002,003,004,012,014,022,023,044

Notes. The raw feasibility, integrality, and recomputed-objective fields are present for all 34 runs; the two gap folders contain identical objectives and verifier values. At 1%, 11 of 34 $\text{Scale}=-1$ objectives differ from the default-base objective by more than twice the stopping tolerance; six of those 11 also fail strict verification, while five pass it. Thus the speedup is real as timing but cannot be classified as a reliable solution improvement. The campaign’s generic `mip_equivalente` field labels every variant named `base` as `BASE`, irrespective of its internal-scaling parameter, so the pass counts above are rebuilt from the raw verifier fields rather than that label. [†]Partial artifact; its descriptive value is not comparable to a complete 34-instance cell.

Table S8: Archived runtime comparison for the operational coupling-stage activation campaign (Gurobi, 1% MIPGap). Solver-time shifted geometric mean (sgm, $s = 1$ s) and descriptive $\text{PAR10}_{\text{avail}}$ are seconds. The accompanying activation diagnostic found zero coupling-stage activations and byte-identical exports for each full/local-only pair; their small runtime differences therefore measure run-to-run variability, not a coupling-stage effect. The coupling-only variant coincides with Base at the exported-model level.

Class	Variant	sgm (s)	$\text{PAR10}_{\text{avail}}$ (s)
Simple	Base	3.73	10.47
	SA-Aug (local+coupling)	2.02	2.42
	SA-Aug, local only	2.01	2.38
	SA-Aug, coupling only	3.74	10.44
	SA-Mat (local+coupling)	2.04	2.43
	SA-Mat, local only	2.06	2.48
	SA-Mat, coupling only	3.73	10.26
Full	Base	82.55	3379.7
	SA-Aug (local+coupling)	71.41	3315.9
	SA-Aug, local only	70.94	3314.4
	SA-Aug, coupling only	83.69	3381.5
	SA-Mat (local+coupling)	78.75	4384.9
	SA-Mat, local only	78.88	4386.1
	SA-Mat, coupling only	90.21	3394.1
SG-Ter-Mer	Base	14.42	565.6
	SA-Aug (local+coupling)	8.55	555.1
	SA-Aug, local only	8.58	555.2
	SA-Aug, coupling only	14.45	565.9
	SA-Mat (local+coupling)	8.96	555.6
	SA-Mat, local only	8.83	555.6
	SA-Mat, coupling only	14.43	565.8

In SG-Ter-Mer one near-limit instance times out under every variant in this independent batch, so $\text{PAR10}_{\text{avail}}$ is dominated by a single 36000-s penalty. In Full, three or four runs time out per variant, so the raw sgm values use different solved subsets and are descriptive. The attribution is the export/activation result: full equals local only, and coupling only equals Base.

Table S9: Direct SA–Flat reliability audit under Gurobi. Each row uses all available records from one dedicated class–tolerance batch; both SA and Flat were rerun in that batch. Timeouts (TO), strict/mixed original-scale verification failures (S/M), and fixed-pool PAR10 are reported for both coverage policies. Each PAR10 cell is ITT/strict/mixed in seconds, on the capped solver-time cost $c^{(k, \text{solver})}$ of the main text; recomputing with the preprocessing-plus-solver cost shifts every cell by less than 0.2 s and changes no ordering.

Gap	Class	Kernel	N	TO SA/Flat	SA fail S/M	Flat fail S/M	SA PAR10 I/S/M	Flat PAR10 I/S/M
1%	Simple	Augmented	204	0/0	12/12	13/12	2.4/2119.8/2119.8	2.4/2296.2/2119.7
	Simple	Matrix-only	204	0/0	19/18	19/18	2.4/3355.1/3178.7	2.4/3355.1/3178.6
	SG-Ter-Mer	Augmented	68	1/1	1/0	2/0	554.5/1083.9/554.5	541.1/1598.9/541.1
	SG-Ter-Mer	Matrix-only	68	1/1	1/0	1/0	555.0/1083.7/555.0	551.6/1080.8/551.6
	Full	Augmented	34	3/3	6/6	3/3	3312.1/9643.7/9643.7	3374.3/6464.9/6464.9
	Full	Matrix-only	34	3/3	2/2	4/4	3420.9/5533.6/5533.6	3430.6/7554.0/7554.0
0.1%	Simple	Augmented	204	1/0	22/18	25/22	211.9/4090.0/3386.8	30.3/4436.5/3908.1
	Simple	Matrix-only	204	1/1	30/26	30/26	203.5/5491.9/4787.2	203.7/5492.1/4787.4
	SG-Ter-Mer	Augmented	68	1/1	1/0	1/0	556.0/1085.4/556.0	542.4/1071.4/542.4
	SG-Ter-Mer	Matrix-only	68	1/1	1/0	1/0	556.5/1085.6/556.5	552.8/1082.0/552.8
	Full	Augmented	30	3/3	2/2	2/1	3951.0/6342.2/6342.2	4004.3/6290.8/5109.3
	Full	Matrix-only	30	4/4	2/1	0/0	5193.8/7585.0/6389.5	5154.8/5154.8/5154.8

Notes. There are no algorithmic or pipeline aborts in these direct batches. Strict penalizes every strict original-scale failure; mixed penalizes only a failure that also lacks a reliable counterpart on the same instance, as defined in the main text. At Full 0.1%, instances 045–048 are absent from both policies; assigning the same penalty to both missing records changes neither their ordering nor any paired contrast. The key reliability sensitivity is visible in Simple/Augmented: the small Flat solver-time advantage at 0.1% reverses under both strict and mixed costs. SG-Ter-Mer/Augmented retains a small descriptive Flat advantage because neither policy has a mixed failure.

Table S10: Magnitude, dependence, and attrition sensitivities for the direct SA–Flat contrasts (exploratory endpoint hierarchy; see the main text). Panel A reports instance-level and operational-family inference for the two central 0.1% augmented contrasts on both endpoints (preprocessing-plus-solver operational endpoint, capped solver-time decomposition), on the shared paired variable $d_i = \log(T_{\text{Flat},i}/T_{\text{SA},i})$, with the Hodges–Lehmann pseudomedian expressed as the typical SA/Flat ratio $\exp(-\hat{\theta}_{\text{HL}})$. Panel B repeats the family analysis for Gurobi Work units. Panel C relocates the path-specific diagnostic from the main timing table because it does not hold the integer solution fixed across policies. Panel D gives the per-contrast attrition and preprocessing-time distributions of the main flat-control table.

Panel A: paired log time ratio and dependence (central augmented 0.1% contrasts)												
Class	Endpoint	n/G	med. SA/Flat	HL SA/Flat	family-bootstrap	95% CI	p_{inst}	q_6	q_{12}	p_{block}	$q_{6,\text{block}}$	$q_{12,\text{block}}$
Simple	pre+solve	204/34	1.0320	1.0359	[1.0274, 1.0418]	3.3×10^{-11}	2.0×10^{-10}	1.3×10^{-10}	5.1×10^{-8}	3.1×10^{-7}	2.1×10^{-7}	
Simple	solver	204/34	1.0093	1.0135	[1.0019, 1.0212]	0.00173	0.01038	0.02075	0.00266	0.01595	0.03190	
SG-Ter-Mer	pre+solve	67/34	1.0473	1.1577	[1.0170, 1.2065]	0.00031	0.00063	0.00063	0.00056	0.00111	0.00117	
SG-Ter-Mer	solver	67/34	1.0159	1.1229	[0.9797, 1.1309]	0.02616	0.05232	0.09346	0.03320	0.06640	0.11980	

Panel B: deterministic effort ($\text{Work}_{\text{Flat}} - \text{Work}_{\text{SA}}$)									
Class	n	Exact zeros	med. Work SA	med. Work Flat	r_{fb}	p_{block}	$q_{6,\text{block}}$	$q_{12,\text{block}}$	
Simple	204	138	2	2	-0.382	0.01196	0.03587	0.04782	
SG-Ter-Mer	67	31	7	4	-0.649	0.000970	0.00582	0.00637	

Panel C: median own-incumbent path-diagnostic ratio					
Gap	Class	Kernel	n	med. $\rho_{\kappa,\text{SA}}^{\text{solver}}/\rho_{\kappa,\text{Flat}}^{\text{solver}}$	
1%	Simple	Augmented	204	1.000	
	Simple	Matrix-only	204	1.000	
	SG-Ter-Mer	Augmented	67	0.626	
	SG-Ter-Mer	Matrix-only	67	0.849	
	Full	Augmented	31	0.628	
	Full	Matrix-only	32	0.849	
0.1%	Simple	Augmented	204	1.000	
	Simple	Matrix-only	203	1.000	
	SG-Ter-Mer	Augmented	67	0.629	
	SG-Ter-Mer	Matrix-only	67	0.849	
	Full	Augmented	27	0.634	
	Full	Matrix-only	27	0.849	

Panel D: per-contrast attrition and preprocessing-time distributions											
Gap	Class	Kernel	Nominal	Recorded	Common pool	Double noncompl.	n used	med. $T_{\text{pre}}^{\text{SA}}$ (s)	med. $T_{\text{pre}}^{\text{Flat}}$ (s)	med. diff (s)	
1%	Simple	Augmented	204	204	204	0	204	0.074	0.025	+0.050	
	Simple	Matrix-only	204	204	204	0	204	0.077	0.025	+0.053	
	SG-Ter-Mer	Augmented	68	68	68	1	67	0.049	0.019	+0.031	
	SG-Ter-Mer	Matrix-only	68	68	68	1	67	0.051	0.019	+0.033	
	Full	Augmented	34	34	34	3	31	0.121	0.041	+0.080	
	Full	Matrix-only	34	34	34	2	32	0.125	0.041	+0.084	
0.1%	Simple	Augmented	204	204	204	0	204	0.077	0.025	+0.051	
	Simple	Matrix-only	204	204	204	1	203	0.079	0.025	+0.054	
	SG-Ter-Mer	Augmented	68	68	68	1	67	0.051	0.019	+0.032	
	SG-Ter-Mer	Matrix-only	68	68	68	1	67	0.052	0.019	+0.033	
	Full	Augmented	34	30	30	3	27	0.120	0.041	+0.079	
	Full	Matrix-only	34	30	30	3	27	0.124	0.041	+0.083	